Ruiwen Wang

Ph.D. Candidate | AI Systems | HPC | ML Systems

Sorbonne University & EURECOM | CIFRE with Huawei Paris Research Center | Expected graduation: end of 2026

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Profile

Ph.D. (CIFRE) candidate in Computer Science at Sorbonne University and EURECOM, in collaboration with Huawei Paris Research Center (expected end of 2026). I design symbolic, optimisation-based planners that turn large-scale hybrid-parallel LLM/DNN training into a predictable, retargetable decision problem — synthesising memory-feasible, communication-aware strategies and pipeline schedules validated on production Ascend clusters up to 16,384 NPUs. Output so far: four first-author papers (CLUSTER, Euro-Par Best Poster, NPC, HPC Asia), one under submission, co-authored large-scale MoE training, and three patent families, several transferred to production.

Doctoral Research

Thesis — *Systematic and Portable Optimisation of Hybrid Parallelism for Large-Scale Distributed Training* (Sorbonne University). A portable, behaviour-level planning workflow built on a single per-layer (Transformer) cost unit that spans all seven parallelism dimensions — data, tensor, pipeline, virtual-pipeline, sequence, expert, and optimiser — and makes large-scale hybrid-parallel planning **predictable** (Prism), **interoperable** (BMPipe + H2O), **evolvable** (Concord), and **adaptable** (ManuMatic). The unit separates model, hardware, and parallelisation descriptions so a plan can be predicted before execution and retargeted across GPU/NPU clusters. Validated across model families — Mixtral-8×7B, Llama3-8B, Qwen2.5-72B, DeepSeek-141B — on Ascend NPU clusters from tens to 16,384 devices, including long-sequence regimes.

Selected Systems & Results

- **Prism — predictability via symbolic memory modeling** (HPC Asia 2026). A profiling-free, layer-level cost surface that predicts per-stage **peak memory** (and communication) analytically from the model, training configuration, and hardware description, turning memory feasibility from trial-and-error into a closed-form, retargetable check across model families and clusters.
- **BMPipe — pipeline efficiency** (IEEE CLUSTER 2025). A symbolic bubble model (recomputation / stage-imbalance / preparation bubbles) coupled with an ILP that jointly sets stage boundaries, intra-stage chunking, and per-layer recomputation to drive down pipeline idle time; up to 1.36× throughput at 83–92% HBM utilization, validated to 16,384 Ascend NPUs.
- **H2O — holistic, all-dimension joint optimisation** (Euro-Par 2025, Best Poster Award). Co-optimises parallelism degrees across all dimensions together with micro-batch size and adaptive recomputation in one workflow, instead of tuning each knob separately; raises MFU 26.13%→35.73% on DeepSeek-141B and transferred to production training.
- **Concord — search efficiency through reuse** (under submission, 2026). Dependency-keyed per-layer signatures pinpoint exactly which planning work a model or configuration edit invalidates, so re-planning reuses the still-valid majority instead of restarting, cutting redundant search during iterative model–strategy co-design.
- **ManuMatic — adaptability to new operators and runtime effects** (NPC 2025). Injects user/runtime-pinned operator shardings as first-class constraints into automatic search, keeping plans robust when new operators or effects unseen by the abstraction appear, and recovering strong plans where pure auto-search degrades (e.g. +30.1% over an expert-tuned plan on Qwen2.5-72B).

Experience

Distributed and Parallel Computing Lab, Huawei Paris Research Center

Paris, France

Ph.D. Candidate (CIFRE) and Research Engineer – LLM Training Systems

2021 – end 2026

- Built cost-model-driven planners for hybrid and pipeline-parallel training: searching stage partitioning, pipeline scheduling, and recomputation strategies.
- Designed symbolic peak-memory and collective-communication cost models for pre-execution feasibility and performance prediction.
- Ran memory and performance analysis and profiling: operator-level profiling, memory and communication footprint analysis, and bottleneck localization to guide throughput and MFU tuning.
- Built debuggable traces and cross-model, cross-scale regression benchmarks supporting reproducible evaluation and configuration guidance.
- Productionized prototypes into deployable planning workflows with engineering and customer teams, driving transfer to production-scale training.

Role progression: Research Apprentice (2021–2022) -> Research Engineer (2022–2023) -> CIFRE Ph.D. Candidate (2023–end 2026).

Production Impact & Collaboration

- Transferred planner-derived strategies to production training: about 1.1× end-to-end on 4,096-device runs and up to 2.37× on long-sequence training at 512 devices.
- Contributed to large-scale TeleChat3-MoE training optimisation in deployment, with feedback loops on real customer workloads and ultra-scale clusters.
- Worked across research, engineering, and customer-facing platform teams; built reproducible benchmarking and debuggable traces that kept planner decisions trustworthy in operation.

Publications

- **Ruiwen Wang**, Chong Li, Thibaut Tachon, Raja Appuswamy, Teng Su. *BMPipe: Bubble-Memory Co-optimization Strategy Planner for Very-Large DNN Training*. IEEE CLUSTER 2025.
- **Ruiwen Wang**, Chong Li, Raja Appuswamy, Yujie Yuan. *H2O: Holistic Hyperparameter Optimization for Large-Scale Deep Neural Network Training*. Euro-Par 2025. **Best Poster Award.**
- **Ruiwen Wang**, Chong Li, Hongxing Wang, Raja Appuswamy, Yujie Yuan. *ManuMatic: Strategy Injection for Robust Automatic Hybrid Parallelism in Distributed DNN Training*. NPC 2025.
- **Ruiwen Wang**, Philippe Fang, Chong Li, Thibaut Tachon, Raja Appuswamy. *PRISM: Profiling-Free Symbolic Memory-Driven Strategy Planner for Large DNN Model Training*. HPC Asia 2026.
- **Ruiwen Wang**, Thibaut Tachon, Chong Li, Raja Appuswamy. *Can Hybrid-Parallel Planning Support Alternating Model–Strategy Design? Dependency-Keyed Per-Layer Primitives for Re-Planning*. Under submission, 2026.
- Zhengdao Yu, **Ruiwen Wang**, Nelson Lossing, Chong Li. *MLLM Pipeline Bubble Modeling for Large-Scale Training*. HPC Asia 2026.
- Xinzhang Liu, Chao Wang, Zhihua Yang, Zhuo Jiang, **Ruiwen Wang**, et al. *Training Report of TeleChat3-MoE*. arXiv:2512.24157, 2025.

Patents

- Chong Li, Pierre Leca, Thibaut Tachon, **Ruiwen Wang**, Haoran Wang. *Devices and methods for generating execution plans for a machine learning model.* (WO 2025/020165 A1)
- Chong Li, **Ruiwen Wang**, Haoran Wang, Fan Yu. *A load balancing method to improve large model performance.*
- Chong Li, Thibaut Tachon, **Ruiwen Wang**, Haoran Wang. *A multi-dimension parallelism configuration method for large language model training.*

Education

Sorbonne University & EURECOM — Paris, France

Ph.D. in Computer Science — 2023 – end 2026

- Thesis: Systematic and Portable Optimisation of Hybrid Parallelism for Large-Scale Distributed Training; supervisors Prof. Raja Appuswamy and Prof. Chong Li.

Sorbonne University — Paris, France

M.Sc. in Computer Science – Algorithms, Software Science and Technology — 2019 – 2022

- Supervisors: Prof. Emmanuel Chailloux and Prof. Jacques Malenfant.

Paris-Saclay University — Paris, France

B.Sc. in Computer Science — 2016 – 2019

- Early research mentored by Prof. Isabelle Guyon, Dr. Viviane Pons, and Dr. Zhengying Liu.

Keywords

Hybrid parallelism · pipeline scheduling · MoE / expert parallelism · symbolic cost & peak-memory modeling · ILP / MILP planning · recomputation · communication overlap · MFU optimisation · Ascend NPU · portability / retargeting.

Awards

- **Golden Prize**, Huawei Challenge Award for Automatic Load Balancing — 2024.08
- **Best Poster Award**, Euro-Par conference — 2025.08
- **Letter of Appreciation**, China Telecom for TeleChat3-MoE deployment — 2025.12

Skills

Parallelism: data, tensor, pipeline, virtual-pipeline, sequence, expert, optimiser

Optimisation: ILP / MILP, constraint programming, symbolic cost & peak-memory modeling

Programming: Python, C++, Java

ML stacks: MindSpore, PyTorch; Ascend NPU / GPU clusters (to 16k), cloud

Tooling: Ascend profiling, tracing, performance regression benchmarking; Linux, Git

Languages: Chinese, English (fluent), French (fluent), Cantonese